

Swati Agrawal

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EDUCATION

University of Massachusetts Amherst, - Master of Science in Computer Science	Dec 2024
Amity University, Noida, India - Bachelors of Technology in Computer Science	May 2019

SKILLS

Programming	Python, SQL (MySQL, SQL Server, Oracle), Java, JavaScript, Ruby & Go; Frameworks: Django, React
Libraries	NumPy, Pandas, Matplotlib, Scikit-Learn, TensorFlow, LangChain
Tools	GitHub, Git, Docker, AWS, Azure; Statistical Analysis, Machine Learning
Security	Penetration Testing, Metasploit, Wireshark, NMAP, Firewalls; Protocols: WAN, LAN, NAT, TCP/IP, SSH, SSL, FTP
Certification	"Solutions Architect – Associate" – AWS*

EXPERIENCE

Graduate Teaching Assistant - Computer and Network Security University of Massachusetts	May 2024 – Present Amherst, MA
- Designed & delivered SQL injection training sessions using Kali Linux & SQL Map, leveraging data analytics to track and optimize student progress. - Increased student enrollment by 30% through engagement analysis and personalized content on Piazza, while creating automated grading and tracking tools in MS Excel and Power BI for accurate performance evaluation.	
Engineer - BI DelveInsight Business Research LLP	Aug 2020 – Nov 2022 Delhi, India
- Created an automated time-tracking tool using Excel and VBA Macros, improving task monitoring efficiency by 25% and providing valuable insights into project progress through automated reports. - Built a dynamic performance dashboard using VBA Macros, reducing manual data collection by 80% and enhancing team productivity by 20% through real-time performance tracking. - Designed data models and interactive dashboards using Power BI, delivering actionable insights into market trends and resulting in a 15% increase in lead conversions by enabling data-driven decision-making. - Streamlined data reporting processes by automating data collection and analysis pipelines, reducing report generation time by 30% and empowering stakeholders to make quicker, data-backed decisions using real-time analytics.	
Analyst Dyson	Nov 2018 – Nov 2019 Gurgaon, India
- Delivered technical support, diagnosing, and troubleshooting Dyson product issues, leading to a 40% increase in customer satisfaction during Dyson's early expansion in India, while significantly enhancing communication and interpersonal skills. - Monitored supply chain operations and managed financial transactions, resulting in a 15% improvement in operational efficiency, supporting the growing customer base.	

PROJECTS

AI Chatbot (Security Lenas) – CVE Finder - OpenAI, LangChain, LLM	GitHub Link
- Developed Django-based AI chatbot that examines Nmap scan results to find system vulnerabilities. It uses the NVD database, OpenAI API, and LangChain to provide CVE details, exploit methods, and defense tips - Enhanced user experience by designing an intuitive web interface with React, while achieving a 40% reduction in setup time through Docker containerization and improving scalability and uptime by 5.2% with AWS Cloud deployment	
Elevation Navigation System for Hikers and Bikers - Java, React, API	GitHub Link
- Designed & Developed a navigation system using Java & React by integrating Google Map API. This system will provide a dynamic UI & real-time mapping and this technique is utilized to add a customized user interface using API. - Enhanced real-time path adjustments and elevations by optimizing the environmental variables from API. Enhanced application security by incorporating API management and user experience.	
Distributed Server App for Stock Trading – DevOps, SDLC	GitHub Link
Designed and developed a distributed server application with microservices for stock trading, implementing gRPC and socket connections for seamless communication between client and server microservices, while adhering to the software development life cycle (SDLC) to ensure high-quality delivery.	
LLM Personalization — A Study with News and Social Media Data - Language Models - FLAN-T5	GitHub Link
Study evaluated FLAN-T5 language model's effectiveness in personalizing content categorization for datasets like news articles & social media posts. By simulating user queries via OpenAI API, study demonstrated 30% improvement in content categorization accuracy & user engagement. This was achieved through fine-tuning, showcasing model's ability to handle diverse language styles & enhance AI-driven content recommendations.	

FREELANCE PROJECT

Ecommerce Website (Full Stack)	Website Link
Developed full-stack e-commerce website for an Indian herbal cosmetic brand using WordPress and Canva. Integrated logistic support features, live package tracking, and a chatbot to enhance user experience and streamline order management. Leading to a 40% increase in product sales.	